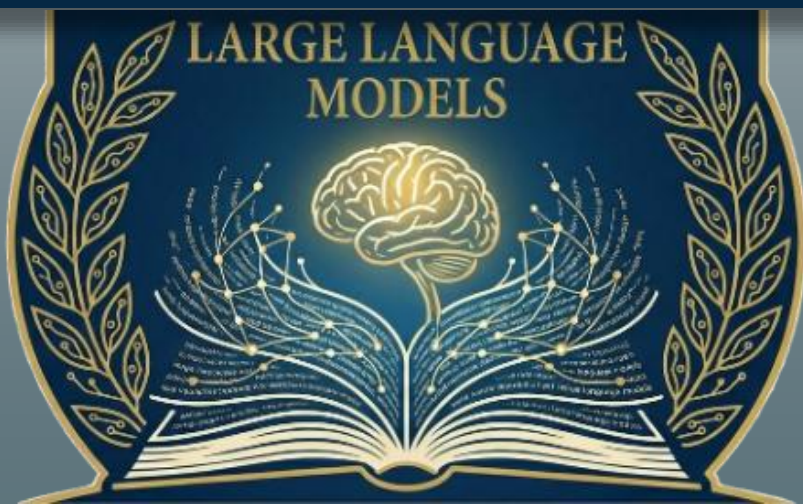


# Fine-Tuning & In-Context Learning



Mohammad Javad Dousti

# Customizing LLMs via Full Model Finetuning

# How we conceived of LMs+NLP several years ago.

- ❑ One model per task trained exclusively on task-specific data.
  - (left) Original transformer paper trained on MT
  - (right) Earliest LM-based chatbot paper trained on movie subtitles

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

| Model                           | BLEU        |              | Training Cost (FLOPs)                 |                     |
|---------------------------------|-------------|--------------|---------------------------------------|---------------------|
|                                 | EN-DE       | EN-FR        | EN-DE                                 | EN-FR               |
| ByteNet [18]                    | 23.75       |              |                                       |                     |
| Deep-Att + PosUnk [39]          |             | 39.2         |                                       | $1.0 \cdot 10^{20}$ |
| GNMT + RL [38]                  | 24.6        | 39.92        | $2.3 \cdot 10^{19}$                   | $1.4 \cdot 10^{20}$ |
| ConvS2S [9]                     | 25.16       | 40.46        | $9.6 \cdot 10^{18}$                   | $1.5 \cdot 10^{20}$ |
| MoE [32]                        | 26.03       | 40.56        | $2.0 \cdot 10^{19}$                   | $1.2 \cdot 10^{20}$ |
| Deep-Att + PosUnk Ensemble [39] |             | 40.4         |                                       | $8.0 \cdot 10^{20}$ |
| GNMT + RL Ensemble [38]         | 26.30       | 41.16        | $1.8 \cdot 10^{20}$                   | $1.1 \cdot 10^{21}$ |
| ConvS2S Ensemble [9]            | 26.36       | <b>41.29</b> | $7.7 \cdot 10^{19}$                   | $1.2 \cdot 10^{21}$ |
| Transformer (base model)        | 27.3        | 38.1         | <b><math>3.3 \cdot 10^{18}</math></b> |                     |
| Transformer (big)               | <b>28.4</b> | <b>41.8</b>  | $2.3 \cdot 10^{19}$                   |                     |

“Attention is all you need.” 2018.

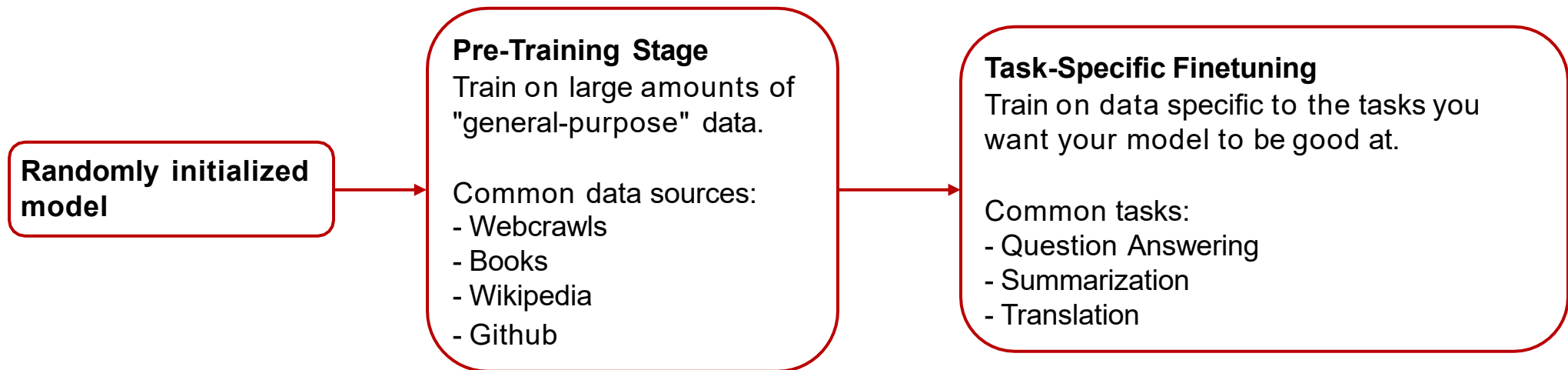
## 4.2. OpenSubtitles dataset

We also tested our model on the OpenSubtitles dataset (Tiedemann, 2009). This dataset consists of movie conversations in XML format. It contains sentences uttered by characters in movies. We applied a simple processing step removing XML tags and obvious non-conversational text (e.g., hyperlinks) from the dataset. As turn taking is not clearly indicated, we treated consecutive sentences assuming they were uttered by different characters. We trained our model to predict the next sentence given the previous one, and we did this for every sentence (noting that this doubles our dataset size, as each sentence is used both for context and as target). Our training and validation split has 62M sentences (923M tokens) as training examples, and the validation set has 26M sentences (395M tokens). The split is done in such a way that each sentence in a pair of sentences either appear together in the training set or test set but not both. Unlike the previous dataset, the OpenSubtitles is quite large, and rather noisy because consecutive sentences may be uttered by the same character. Given the broad scope of movies, this is an open-domain conversation dataset, contrasting with the technical troubleshooting dataset.

“A Neural Conversational Model” (2015) <https://arxiv.org/abs/1506.05869>

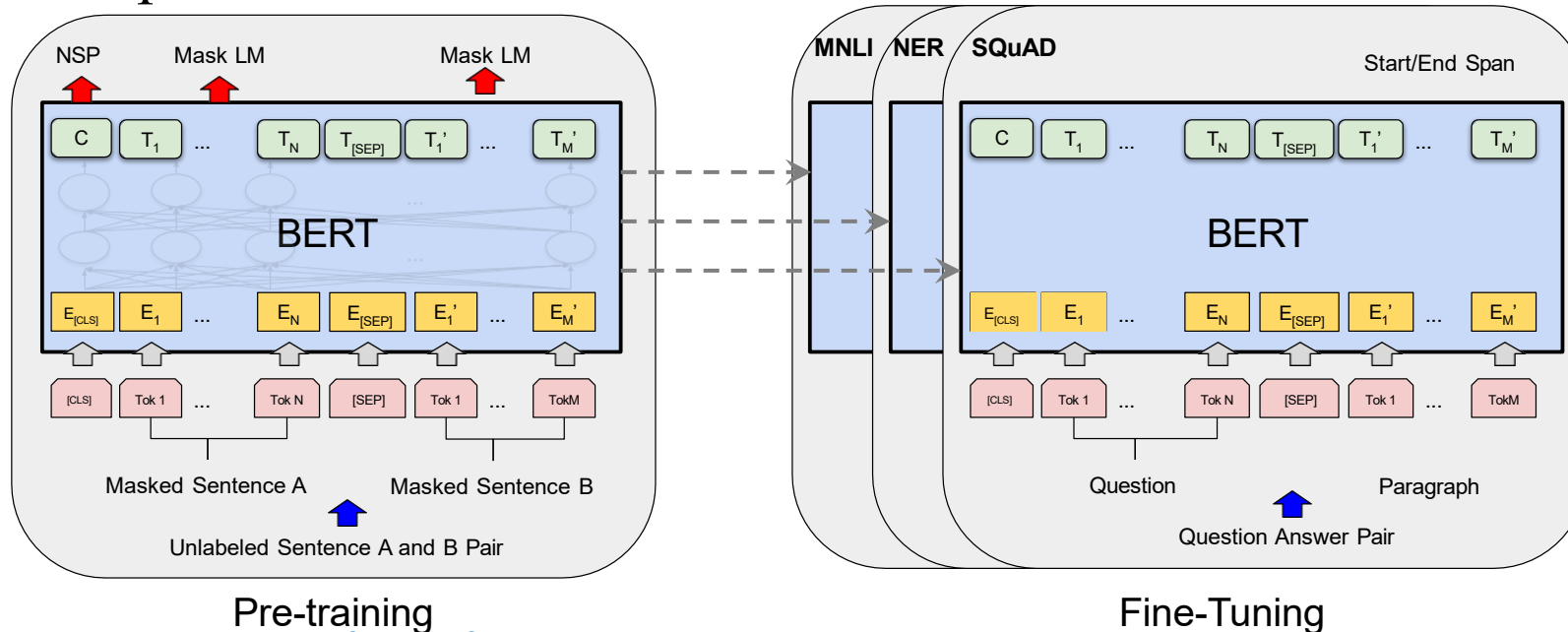
# Paradigm shift around ~2018

- ❑ One language model pre-trained for generic language understanding then finetuned for each task.



# Paradigm shift around ~2018

- ❑ One language model pre-trained for generic language understanding then finetuned for each task.
- ❑ BERT:
  - Pre-train an encoder-only model with mask infilling and sentence ordering objectives.
  - Finetune once per NLP task



Source: ["BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." 2018.](#)

# Paradigm shift around ~2018

- ❑ One language model pre-trained for generic language understanding then finetuned for each task.
- ❑ GPT-1:
  - Pre-train a decoder-only LM with a language modelling objective.
  - Finetune once per NLP task

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

| Method                              | MNLI-m      | MNLI-mm     | SNLI        | SciTail     | QNLI        | RTE         |
|-------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| ESIM + ELMo [44] (5x)               | -           | -           | <u>89.3</u> | -           | -           | -           |
| CAFE [58] (5x)                      | 80.2        | 79.0        | <u>89.3</u> | -           | -           | -           |
| Stochastic Answer Network [35] (3x) | <u>80.6</u> | <u>80.1</u> | -           | -           | -           | -           |
| CAFE [58]                           | 78.7        | 77.9        | 88.5        | <u>83.3</u> |             |             |
| GenSen [64]                         | 71.4        | 71.3        | -           | -           | <u>82.3</u> | 59.2        |
| Multi-task BiLSTM + Attn [64]       | 72.2        | 72.1        | -           | -           | 82.1        | <b>61.7</b> |
| Finetuned Transformer LM (ours)     | <b>82.1</b> | <b>81.4</b> | <b>89.9</b> | <b>88.3</b> | <b>88.1</b> | 56.0        |

Source: [“Improving Language Understanding by Generative Pre-Training.” 2018.](#)

# Disadvantages of full model-finetuning

- ❑ One model per task is fine for small models, but not for today's big ones.
- ❑ Training separately on each task is more expensive than training on all tasks at once.
- ❑ Overfitting / catastrophic forgetting on small datasets
- ❑ Storage of all those model checkpoints (one per task) is expensive.
- ❑ During inference time, it's expensive to have all those models loaded onto GPUs at the same time.

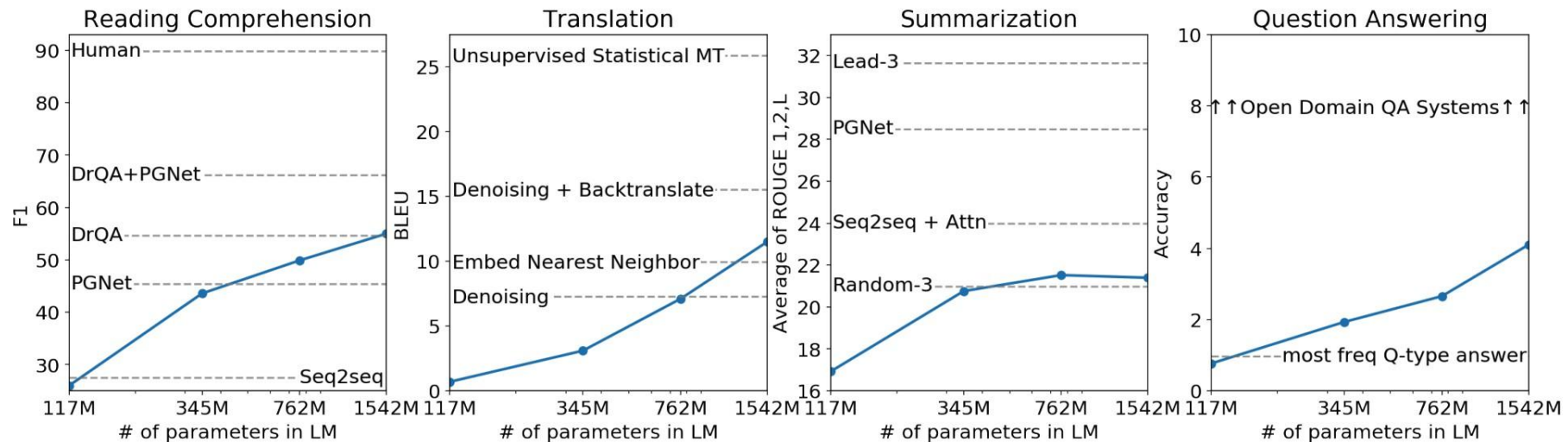
# Solutions to Improve Efficiency

- ❑ Avoid finetuning entirely
  - In-context learning
- ❑ Parameter-efficient finetuning
- ❑ Multi-task finetuning → instruction finetuning

# In-Context Learning

# Language Models are Unsupervised Multitask Learners (GPT2)

- ❑ The authors “demonstrate that language models begin to learn [question answering, machine translation, reading comprehension, and summarization] tasks **without any explicit supervision** when trained on a new dataset of millions of webpages called WebText.”
- ❑ Across a variety of tasks, they saw this behavior “emerge” with larger model sizes.



Source: ["Language Models are Unsupervised Multitask Learners." Radford et al. 2019.](#)

# In-Context learning (1)

- ❑ In-context learning was popularized in the original GPT-3 paper as a way to use language models to learn tasks given only a few examples.
- ❑ On many benchmark NLP benchmarks, in-context learning is competitive with models trained with much more labeled data.
- ❑ **Few-shot prompting** can be used as a technique to enable **in-context learning** where we provide demonstrations in the prompt to steer the model to better performance.



Source: " [Language Models are Few-Shot Learners.](#)" Brown et al. 2020.

# In-Context Learning (2)

- ❑ All the information necessary to get the LM to do the task is included as part of the textual prompt inputted to the LM.
- ❑ **LLM zero-shot learning:** A prompt that contains instructions for the task, but no actual examples of the task being performed.
- ❑ **LLM few-shot learning:** A prompt that contains both instructions as well as several examples of the task being performed.

# Few-shot Learning with LLMs

- Suppose you have...
  - a dataset  $D = \{(x_i, y_i)\}_{i=1}^N$  and  $N$  is rather small (i.e. few-shot setting)
  - a very large (billions of parameters) pre-trained language model
- There are two ways to “learn”:

Our focus



## Option A: Supervised fine-tuning

- **Definition:** fine-tune the LLM on the training data using...
  - a standard supervised objective
  - backpropagation to compute gradients
  - your favorite optimizer (e.g. Adam)
- **Pro:** fits into the standard ML recipe
- **Pro:** still works if  $N$  is large
- **Con:** backpropagation requires  $\sim 3x$  the memory and computation time as the forward computation
- **Con:** you might not have access to the model weights at all (e.g. because the model is proprietary)

## Option B: In-context learning

- **Definition:**
  - feed training examples to the LLM as a prompt
  - allow the LLM to infer patterns in the training examples during inference (i.e. decoding)
  - take the output of the LLM following the prompt as its prediction
- **Con:** the prompt may be very long and Transformer LMs require  $O(N^2)$  time/space where  $N$  = length of context
- **Pro:** no backpropagation required and only one pass through the training data
- **Pro:** does not require model weights, only API access

# Zero-shot learning for sentiment classification

## Prompt:

Review: Let there be no question: Alexions owns the best cheeseburger in the region and they have now for decades. Try a burger on Italian bread. The service is flawlessly friendly, the food is amazing, and the wings? Oh the wings... but it's still about the cheeseburger. The atmosphere is inviting, but you can't eat atmosphere... so go right now. Grab the car keys... you know you're hungry for an amazing cheeseburger, maybe some wings, and a cold beer! Easily, hands down, the best bar and grill in Pittsburgh.

On a 1 to 4 star scale, the reviewer would probably give this restaurant a

## Model's Output Probabilities:

$P("1" \mid \text{prompt}) = 0.0016$

$P("2" \mid \text{prompt}) = 0.0032$

$P("3" \mid \text{prompt}) = 0.012$

$P("4" \mid \text{prompt}) = 0.78$

## Template used:

Review: `<review>`

On a 1 to 4 star scale, the reviewer would probably give this restaurant a  
`<class_label>`

This template is tried for all possible class labels.

Output: the number with the highest likelihood.

# Zero-shot learning for abstractive summarization

## Prompt:

India's moon rover completes its walk. Scientists analyzing data looking for signs of frozen water BEW DELHI -- India's moon rover has completed its walk on the lunar surface and been put into sleep mode less than two weeks after its historic landing near the lunar south pole, India's space mission said. "The rover completes its assignments. It is now safely parked and set into sleep mode," with daylight on that part of the moon coming to an end, the Indian Space Research Organization said in a statement late Saturday. The rover's payloads are turned off and the data it collected has been transmitted to the Earth via the lander, the statement said.

[...]

Active since the 1960s, India has launched satellites for itself and other countries, and successfully put one in orbit around Mars in 2014. India is planning its first mission to the International Space Station next year, in collaboration with the United States.

TL;DR;

## Model Generation:

India's moon rover has completed its assignments and gone to sleep mode after just two weeks of being on the lunar surface. The rover has started transmitting the data it has collected to Earth, but scientists are unsure if it can survive the extremely cold temperatures on the moon and wake up in order to conduct another set of missions when the sun rises again. India successfully landed the rover and underscored its status as a major tech power and space program.

## Template used:

```
<title>  
<article>  
TL;DR;
```

We return the model's generation from the filled out template.

# Few-shot In-Context Learning (example 1)

- LLMs can solve novel downstream tasks by just conditioning on a few demonstrations of the task in its prefix.

Circulation revenue has increased by 5% in Finland. // Positive

Panostaja did not disclose the purchase price. // Neutral

Paying off the national debt will be extremely painful. // Negative

The company anticipated its operating profit to improve. // \_\_\_\_\_

LM

Circulation revenue has increased by 5% in Finland. // Finance

They defeated ... in the NFC Championship Game. // Sports

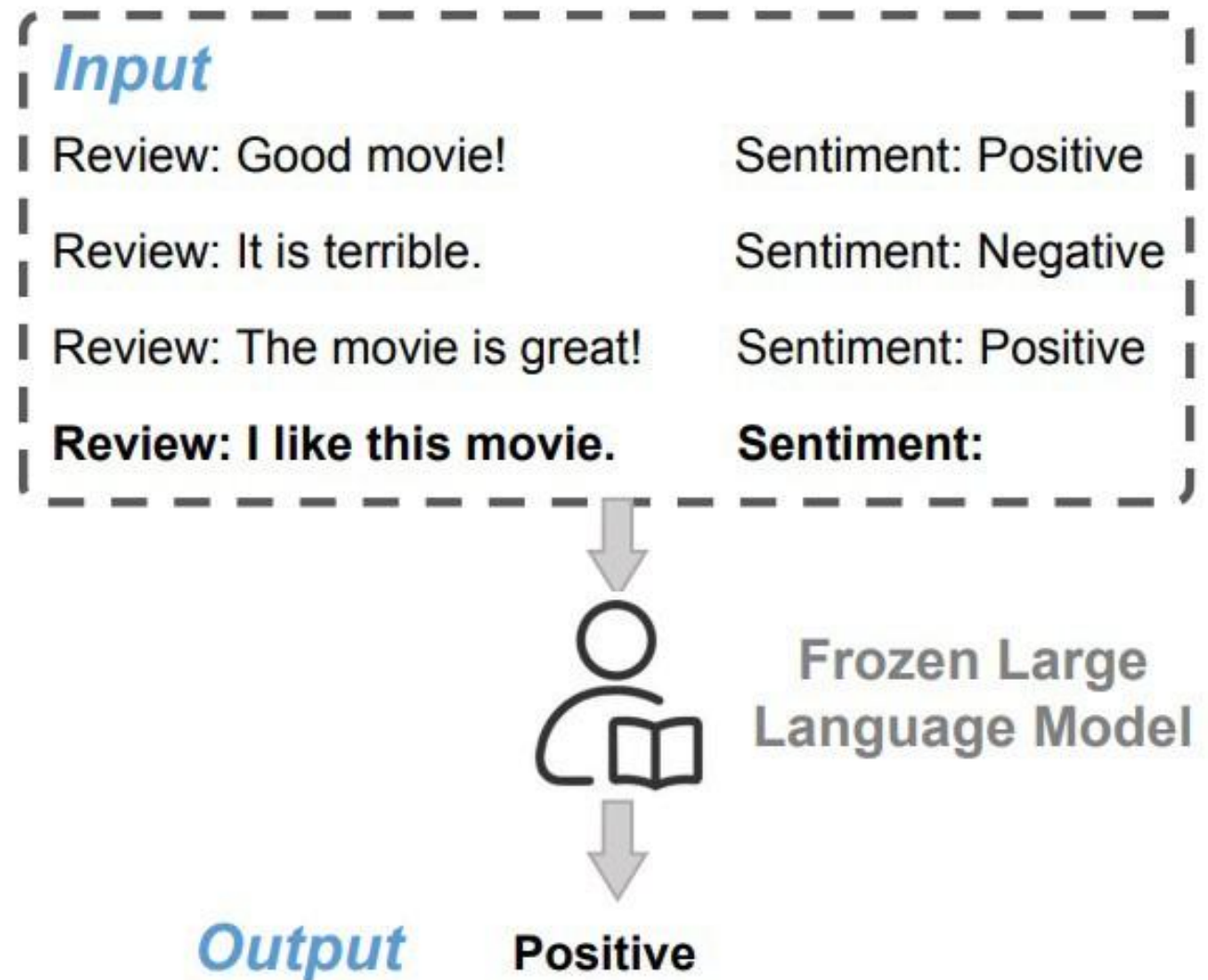
Apple ... development of in-house chips. // Tech

The company anticipated its operating profit to improve. // \_\_\_\_\_

LM

# Few-shot In-Context Learning (example 2)

- ❑ Few-shot learning can be done via in-context learning
- ❑ Typically, a task description is presented first
- ❑ Then a sequence of input/output pairs from a training dataset are presented in sequence



# Few-shot In-Context Learning (example 3)

- ❑ Few-shot learning can be done via in- context learning
- ❑ Typically, a task description is presented first
- ❑ Then a sequence of input/output pairs from a training dataset are presented in sequence

The three settings we explore for in-context learning

## Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



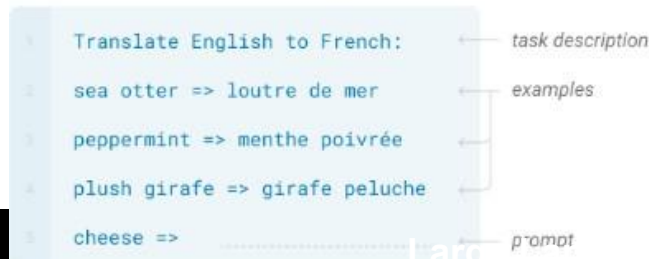
## One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.



## Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Traditional fine-tuning (not used for GPT-3)

## Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



# Few-shot In-Context Learning (example 3: MT)

## Prompt:

The dog chased a squirrel at the park. = 那只狗在公园里追一只松鼠。  
I was late for class. = 我上□□到了。  
The hippopotamus ate my homework. =

## Model Generation:

河□吃了我的家庭作□。

## Template Used:

<example1\_en> = <example1\_zh>

<example2\_en> = <example2\_zh>

<query\_en> =

# There are often many ways to verbalize a task.

## Prompt:

The dog chased a squirrel at the park. = 那只狗在公园里追一只松鼠。  
I was late for class. = 我上□□到了。  
The hippopotamus ate my homework. =

## Prompt with an Alternative Template:

Translate from English to Chinese.  
The dog chased a squirrel at the park. = 那只狗在公园里追一只松鼠。  
  
I was late for class. = 我上□□到了。  
  
The hippopotamus ate my homework. =

## Prompt with an Alternative Template:

Translate from English to Chinese.

English: The dog chased a squirrel at the park.  
Chinese: 那只狗在公园里追一只松鼠。

English: I was late for class.  
Chinese: 我上□□到了。

English: The hippopotamus ate my homework.  
Chinese:

The different templates are called **verbalizers**.

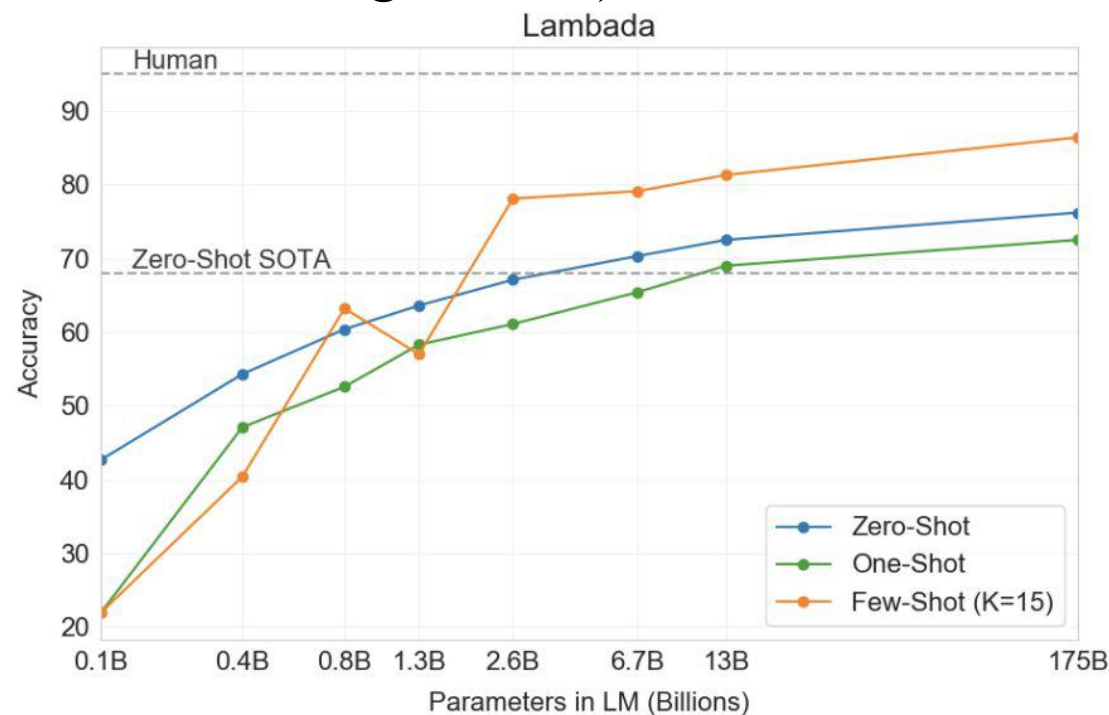
# Key Terms Definition

- ❑ **Emergence:** When quantitative changes in a system result in qualitative changes in behavior.
- ❑ **Emergent behaviors:** Abilities that larger models have and smaller models don't
- ❑ **In-context learning:** When a language model “learns” how to do a task from a textual prompt containing a natural language instruction for the task, several exemplars of the task, or both.
- ❑ **Zero-shot learning:** In context learning that does not include any exemplars of the task.
- ❑ **Few-shot learning:** In context learning that contains several exemplars of the task.
- ❑ **Prompt engineering:** The painstaking process of trying out many different prompts until you find one that works well for your task.
- ❑ **Verbalizer:** The template we wrap an example in in order to perform the task.

- ❑ More on prompt engineering (why it works, cases where it doesn't) in near future.

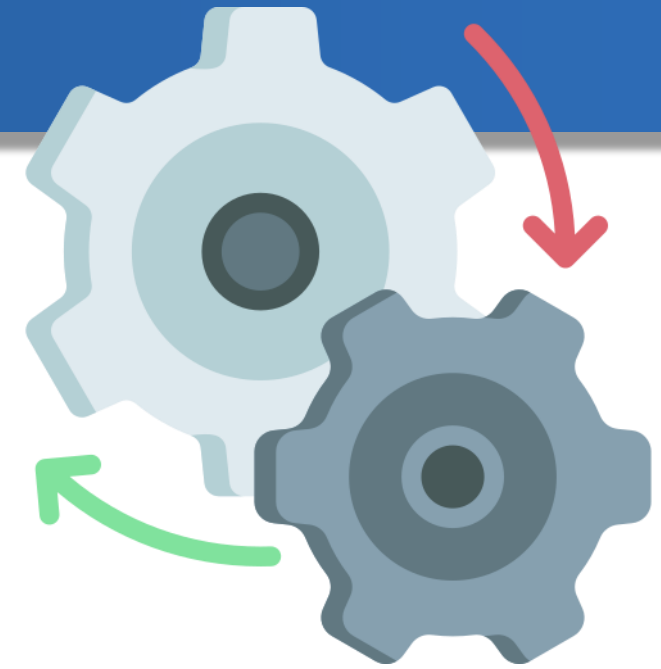
# Why is in-context learning surprising?

- ❑ Unlike conventional machine learning: No optimization of any parameters.
  - This isn't unique: Meta-learning methods have trained models that learn from examples.
- ❑ The mystery is that the LM isn't trained to learn from examples.
- ❑ There's seemingly a mismatch between pretraining (what it's trained to do, which is next token prediction) and in-context learning (what we're asking it to do).
- ❑ This seems like magic!

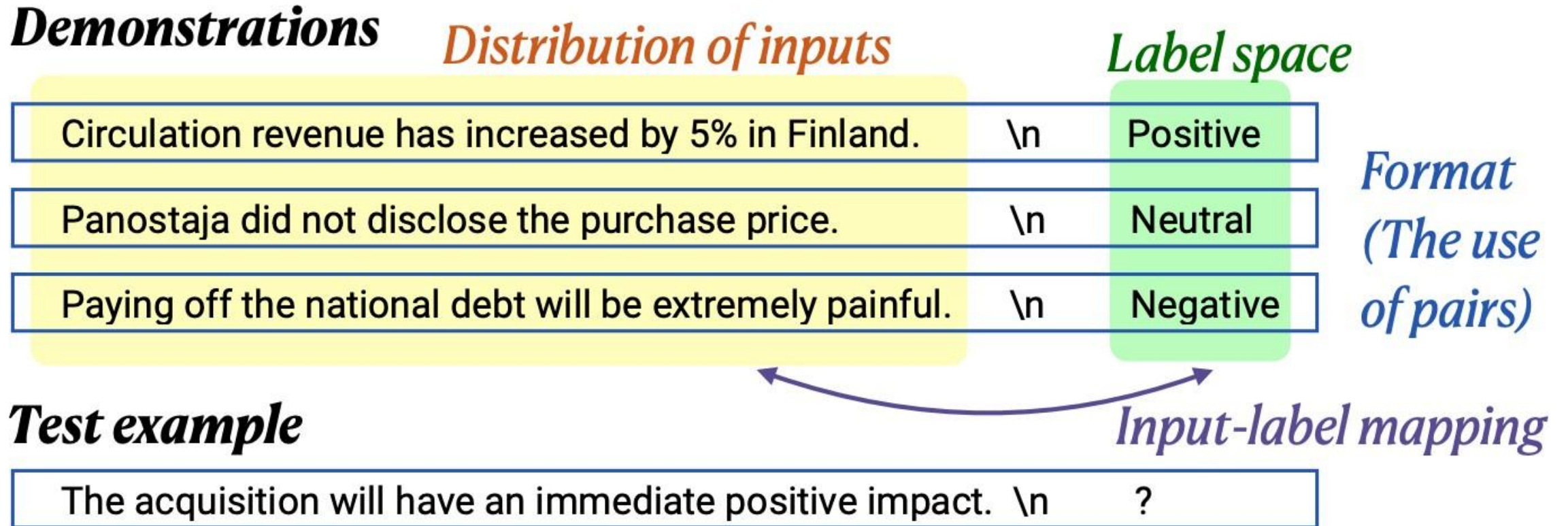


# How does this work?

- ❑ Are “**novel**” tasks really novel? Or are they buried somewhere within the trillions of words in the Common Crawl?
- ❑ If not, how can the model “**learn**” to solve a new task without any training (i.e., gradient updates to optimize model performance on the new task)?
- ❑ In-context learning can be sensitive to...
  - the order the training examples are presented
  - the balance of labels (e.g. positive vs. negative)
  - the number of unique labels covered
  - ... but is it actually?

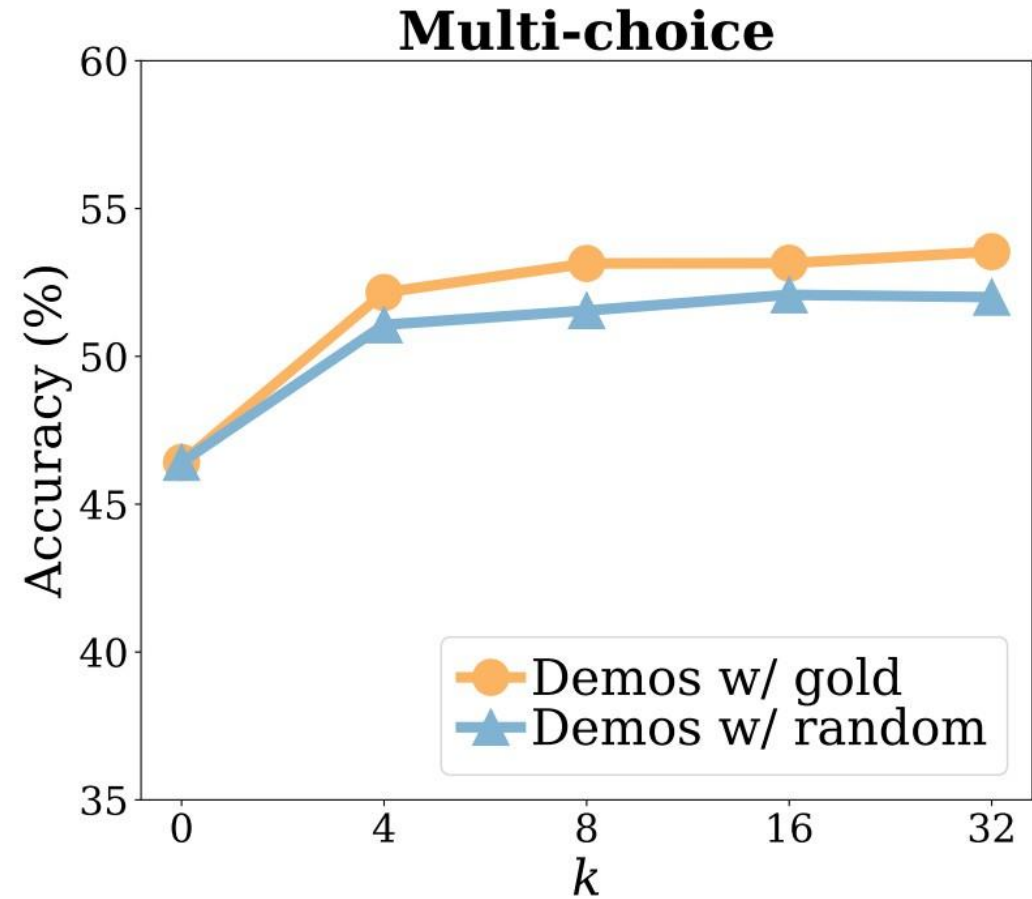
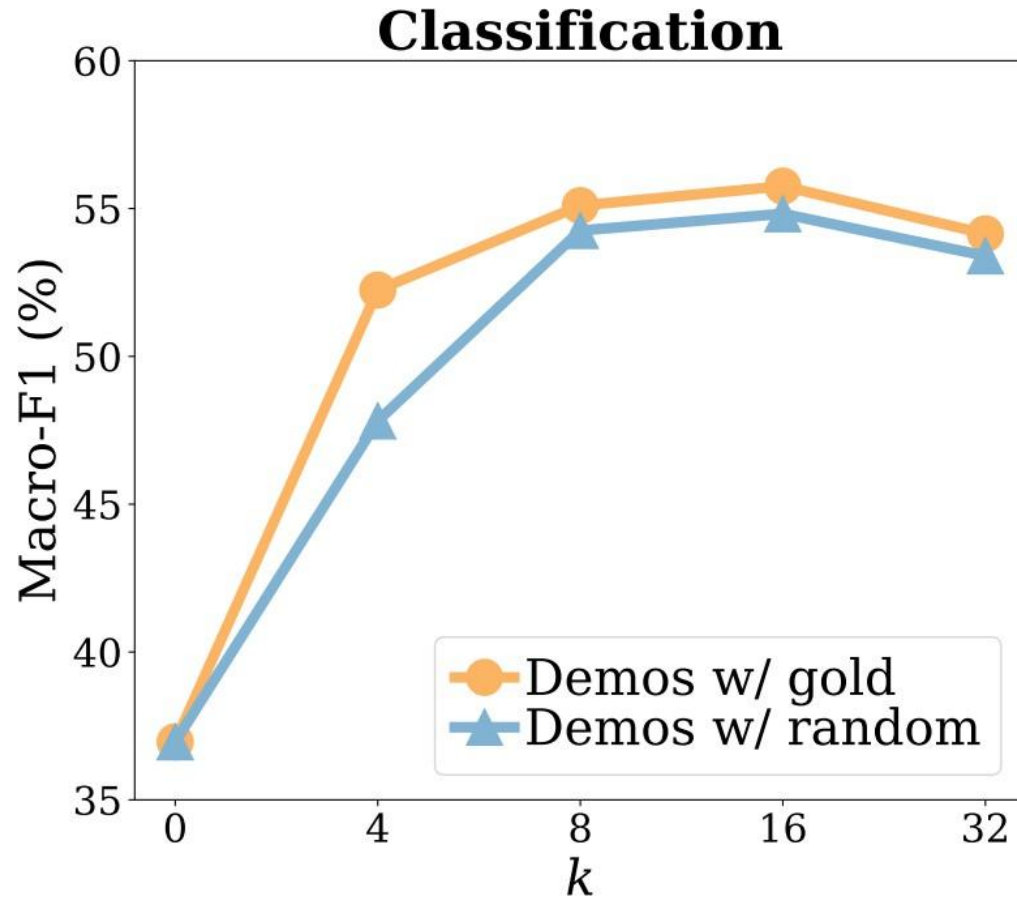


# What's in a demonstration?



Source: [Min et al., "Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?" EMNLP 2022](#)

# How important is input-label mapping?



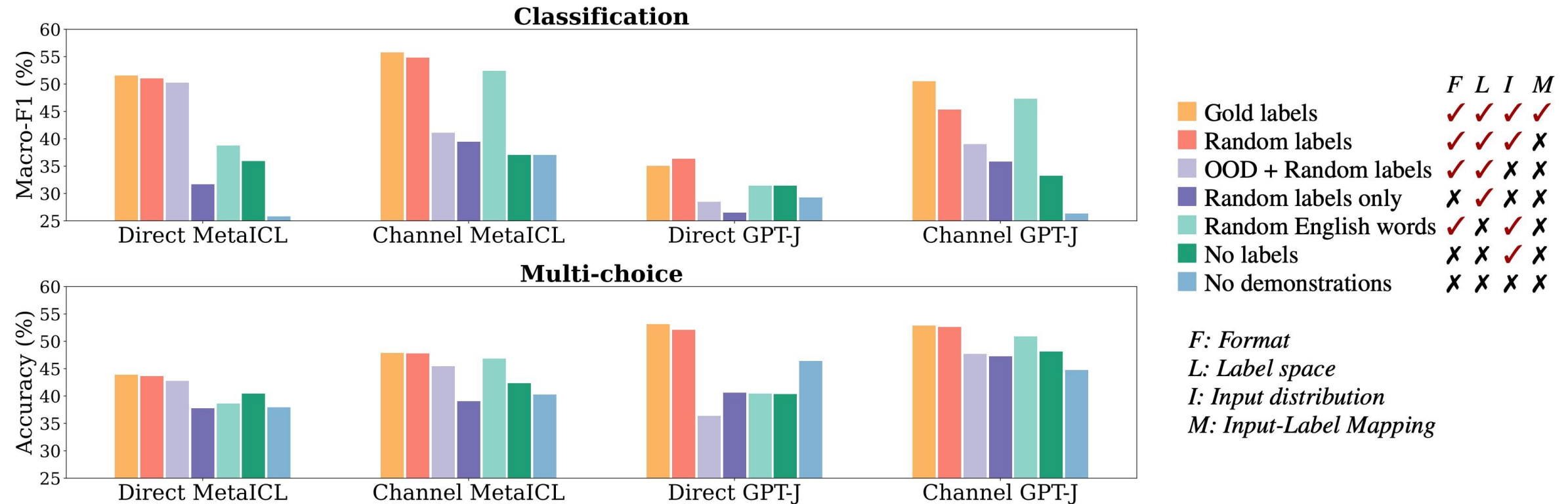
Source: [Min et al., "Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?" EMNLP 2022](#)

# What about other aspects of the demonstrations?

|                                          |                                                                                                                                                                                                                                                                  |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demos<br/>w/ gold labels</i>          | (Format ✓ Input distribution ✓ Label space ✓ Input-label mapping ✓)<br>Circulation revenue has increased by 5% in Finland and 4% in Sweden in 2008. \n positive<br>Panostaja did not disclose the purchase price. \n neutral                                     |
| <i>Demos<br/>w/ random labels</i>        | (Format ✓ Input distribution ✓ Label space ✓ Input-label mapping ✗)<br>Circulation revenue has increased by 5% in Finland and 4% in Sweden in 2008. \n neutral<br>Panostaja did not disclose the purchase price. \n negative                                     |
| <i>OOD Demos<br/>w/ random labels</i>    | (Format ✓ Input distribution ✗ Label space ✓ Input-label mapping ✗)<br>Colour-printed lithograph. Very good condition. Image size: 15 x 23 1/2 inches. \n neutral<br>Many accompanying marketing claims of cannabis products are often well-meaning. \n negative |
| <i>Demos<br/>w/ random English words</i> | (Format ✓ Input distribution ✓ Label space ✗ Input-label mapping ✗)<br>Circulation revenue has increased by 5% in Finland and 4% in Sweden in 2008. \n unanimity<br>Panostaja did not disclose the purchase price. \n wave                                       |
| <i>Demos<br/>w/o labels</i>              | (Format ✗ Input distribution ✓ Label space ✗ Input-label mapping ✗)<br>Circulation revenue has increased by 5% in Finland and 4% in Sweden in 2008.<br>Panostaja did not disclose the purchase price.                                                            |
| <i>Demos<br/>labels only</i>             | (Format ✗ Input distribution ✗ Label space ✓ Input-label mapping ✗)<br>positive<br>neutral                                                                                                                                                                       |

Source: [Min et al., “Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?” EMNLP 2022](#)

# What about other aspects of the demonstrations?



Source: [Min et al., “Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?” EMNLP 2022](#)

# Justification

- ❑ In-context learning can be thought of as “**locating**” **latent concepts** the LM has acquired from pretraining data.
- ❑ All components of the prompt (inputs, outputs, formatting, and the input-output mapping) can provide information for inferring the latent concept.

Circulation revenue has increased by 5%  
in Finland. // Positive

Panostaja did not disclose the purchase  
price. // Neutral

Paying off the national debt will be  
extremely painful. // Negative

The company anticipated its operating  
profit to improve. // \_\_\_\_\_

LM

Circulation revenue has increased by  
5% in Finland. // Finance

They defeated ... in the NFC  
Championship Game. // Sports

Apple ... development of in-house  
chips. // Tech

The company anticipated its operating  
profit to improve. // \_\_\_\_\_

LM

# Justification (cont'd)

- ❑ **Pretrain:** To predict the next token during pre-training, the LM must infer the **latent concept** for the document using evidence from the previous sentences.
- ❑ **In-context learning:** If the LM also **infers the prompt concept** using demonstrations in the prompt, then in-context learning succeeds!

# This is a nice study, but it is also limited

- ❑ Only studied simple classification tasks.
- ❑ The effect of these different aspects of demonstrations on generation tasks is unclear.

# Follow up work tells a different story

- ❑ [Yoo et al. \(2022\)](#) shows that input-label mappings matter quite significantly when using different experimental conditions and eval metrics.
- ❑ [Madaan and Yazdanbakhsh \(2022\)](#) show that random rationales degrade chain-of-thought performance, but other modifications to the rationale (e.g., wrong equations) don't affect it too much.

# How does the LLM make use of (any aspect of) the demonstration?

- “We investigate the hypothesis that transformer-based in-context learners implement standard learning algorithms implicitly, by encoding smaller models in their activations, and updating these implicit models as new examples appear in the context.” — [Akyürek et al., ICLR 2023](#)

# Can a Transformer do linear regression?

- Akyürek et al., ICLR 2023: theoretically proved that LLMs can learn a small linear model:

$$\sum_i \mathcal{L}(\mathbf{w}^\top \mathbf{x}_i, y_i) + \lambda \|\mathbf{w}\|_2^2$$

- The above linear regression loss has a *closed form* solution:

$$\mathbf{w}^* = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$$

- But can also be solved via gradient descent

$$\mathbf{w}' = \mathbf{w} - \alpha \frac{\partial}{\partial \mathbf{w}} \left( \mathcal{L}(\mathbf{w}^\top \mathbf{x}_i, y_i) + \lambda \|\mathbf{w}\|_2^2 \right) = \mathbf{w} - 2\alpha(\mathbf{x}\mathbf{w}^\top \mathbf{x} - y\mathbf{x} + \lambda\mathbf{w})$$

## □ Transformers can implement some primitive operations useful for learning

It will be useful to first establish a few computational primitives with simple transformer implementations. Consider the following four functions from  $\mathbb{R}^{H \times T} \rightarrow \mathbb{R}^{H \times T}$ :

**mov**( $H; s, t, i, j, i', j'$ ): selects the entries of the  $s^{\text{th}}$  column of  $H$  between rows  $i$  and  $j$ , and copies them into the  $t^{\text{th}}$  column ( $t \geq s$ ) of  $H$  between rows  $i'$  and  $j'$ , yielding the matrix:

$$\left[ \begin{array}{c|cc} & H_{:,i-1,t} & \\ H_{:,s:t} & H_{i':j',s} & H_{:,t+1:} \\ & H_{j,t} & \end{array} \right].$$

**mul**( $H; a, b, c, (i, j), (i', j'), (i'', j'')$ ): in *each* column  $\mathbf{h}$  of  $H$ , interprets the entries between  $i$  and  $j$  as an  $a \times b$  matrix  $A_1$ , and the entries between  $i'$  and  $j'$  as a  $b \times c$  matrix  $A_2$ , multiplies these matrices together, and stores the result between rows  $i''$  and  $j''$ , yielding a matrix in which each column has the form  $[\mathbf{h}_{:,i''-1}, A_1 A_2, \mathbf{h}_{j'':}]^\top$ .

**div**( $H; (i, j), i', (i'', j'')$ ): in each column  $\mathbf{h}$  of  $H$ , divides the entries between  $i$  and  $j$  by the absolute value of the entry at  $i'$ , and stores the result between rows  $i''$  and  $j''$ , yielding a matrix in which every column has the form  $[\mathbf{h}_{:,i''-1}, \mathbf{h}_{i:j} / |\mathbf{h}_{i'}|, \mathbf{h}_{j'':}]^\top$ .

**aff**( $H; (i, j), (i', j'), (i'', j''), W_1, W_2, b$ ): in each column  $\mathbf{h}$  of  $H$ , applies an affine transformation to the entries between  $i$  and  $j$  and  $i'$  and  $j'$ , then stores the result between rows  $i''$  and  $j''$ , yielding a matrix in which every column has the form  $[\mathbf{h}_{:,i''-1}, W_1 \mathbf{h}_{i:j} + W_2 \mathbf{h}_{i':j'} + b, \mathbf{h}_{j'':}]^\top$ .

**Lemma 1.** *Each of mov, mul, div and aff can be implemented by a single transformer decoder layer: in Eq. (1) and Eq. (4), there exist matrices  $W^Q, W^K, W^V, W^F, W_1$  and  $W_2$  such that, given a matrix  $H$  as input, the layer's output has the form of the corresponding function output above.*<sup>1</sup>

- Using these primitives, they show that Transformers can implement one step of gradient descent

**Theorem 1.** *A transformer can compute Eq. (11) (i.e. the prediction resulting from single step of gradient descent on an in-context example) with constant number of layers and  $O(d)$  hidden space, where  $d$  is the problem dimension of the input  $x$ . Specifically, there exist transformer parameters  $\theta$  such that, given an input matrix of the form:*

$$H^{(0)} = \begin{bmatrix} \cdots & 0 & y_i & 0 & \cdots \\ & \mathbf{x}_i & 0 & \mathbf{x}_n & \end{bmatrix}, \quad (12)$$

*the transformer's output matrix  $H^{(L)}$  contains an entry equal to  $\mathbf{w}'^\top \mathbf{x}_n$  (Eq. (11)) at the column index where  $\mathbf{x}_n$  is input.*

# Does ICL encode meaningful intermediate quantities?

- Probe networks show that Transformers encode useful intermediate computations while training the linear models.

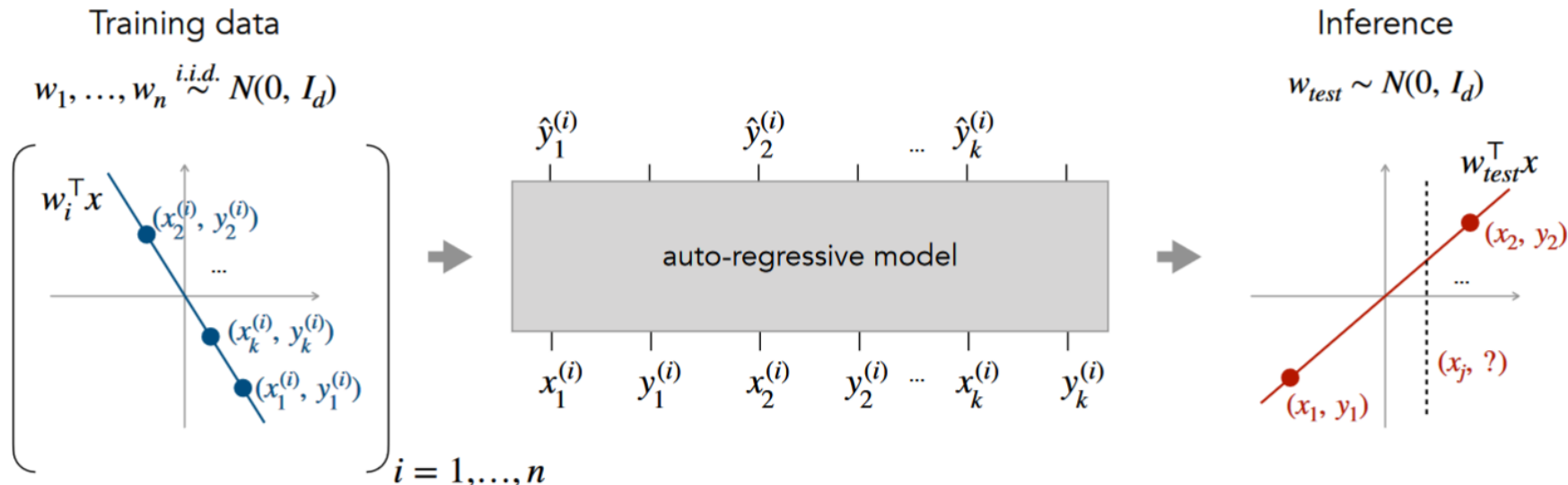
To do so, we identify two intermediate quantities that we expect to be computed by gradient descent and ridge-regression variants: the **moment vector**  $X^\top Y$  and the (min-norm) least-square estimated **weight vector**  $w_{\text{OLS}}$ , each calculated after feeding  $n$  exemplars. We take a trained in-context learner, freeze its weights, then train an auxiliary **probing** model (Alain & Bengio, 2016) to attempt to recover the target quantities from the learner's hidden representations. Specifically, the probe model takes hidden states at a layer  $H^{(l)}$  as input, then outputs the prediction for target variable. We define a probe with **position-attention** that computes (Appendix E):

$$\alpha = \text{softmax}(s_v) \tag{20}$$

$$\hat{v} = \text{FF}_v(\alpha^\top W_v H^{(l)}) \tag{21}$$

# Learn More Functions In-context

- Can we train a model that in-context learns a function class?
  - Transformers can be trained from scratch to perform in-context learning of **unseen linear functions** with performance comparable to the optimal least squares estimator.
  - Transformers can also in-context learn more complex function classes—**namely sparse linear functions, two-layer neural networks, and decision trees**—with performance that matches or exceeds task-specific learning algorithms.



Source: [Garg et al., "What Can Transformers Learn In-Context? A Case Study of Simple Function Classes," NeurIPS 2022.](#)